Gerasimos Kouniatalis

Publication List

gkouniatalis@noa.gr | ORCID 0009-0004-5593-3614 | INSPIRE-HEP

Peer-Reviewed Publications

1. **G. Kouniatalis**, "A Unified Scalar-Field Mechanism for the H_0, S_8 and Evolving Dark-Energy Tensions," *Astronomy* **5**, 17 (2026). doi:10.3390/astronomy5040017; arXiv:2603.11706.
2. **G. Kouniatalis**, T. Papanikolaou, S. Basilakos, E. N. Saridakis, "Primordial black hole formation from transient $f(T)$ cosmology," *Physics of the Dark Universe* **53**, 102418 (2026). doi:10.1016/j.dark.2026.102418; arXiv:2603.24759.
3. S. Basilakos, **G. Kouniatalis**, E. N. Saridakis, C. Tzerefos, "Bohmian quantum cosmology from the Wheeler–DeWitt equation," *Physics Letters B* **875**, 140339 (2026). doi:10.1016/j.physletb.2026.140339; arXiv:2512.18818.
4. **G. Kouniatalis**, A. G. Suvorov, K. Destounis, "Lensing by black holes within astrophysical environments," *Physical Review D* **112**, 124062 (2025). doi:10.1103/tn1g-7tfj; arXiv:2508.19333.
5. **G. Kouniatalis**, P. A. González, E. Papantonopoulos, Y. Vásquez, "Near-horizon geodesic instabilities and anomalous decay of quasinormal modes in Weyl black holes," *Physical Review D* **112**, 124061 (2025). doi:10.1103/ftvr-pclv; arXiv:2509.05164.
6. **G. Kouniatalis**, E. N. Saridakis, "Inflation from a generalized exponential plateau: towards extra suppressed tensor-to-scalar ratios," *Journal of Cosmology and Astroparticle Physics* **11**, 038 (2025). doi:10.1088/1475-7516/2025/11/038; arXiv:2507.17721.
7. E. Di Valentino et al. (including **G. Kouniatalis**), "The CosmoVerse White Paper: Addressing observational tensions in cosmology with systematics and fundamental physics," *Physics of the Dark Universe* **49**, 101965 (2025). doi:10.1016/j.dark.2025.101965; arXiv:2504.01669.
8. **G. Kouniatalis**, E. N. Saridakis, "Modified gravity from Weyl connection and the $f(R, \mathcal{A})$ extension," *European Physical Journal C* **85**, 260 (2025). doi:10.1140/epjc/s10052-025-14017-z; arXiv:2411.14380.

Preprints

9. **G. Kouniatalis**, "Initial conditions for inflation," arXiv:2609.09200 (2026).
10. **G. Kouniatalis**, E. N. Saridakis, "A null test of the Hubble tension," arXiv:2601.22043 (2026).
11. **G. Kouniatalis**, "The Wave Function of the Universe and Inflation," arXiv:2510.04775 (2025; revised 2026).

Author name is shown in bold. Publication status and bibliographic metadata updated through 9 October 2026.